

MIT OpenCourseWare

2.14

Analysis and Design of Feedback Control Systems 2.14 — Analysis and Design of

Feedback Control Systems May 2026 **EXTREMELY HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (4 points) Construct an original proof-level problem involving Transfer Functions for 2.14 (Analysis and Design of Feedback Control Systems) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
2. (4 points) Construct an original proof-level problem involving PID Control for 2.14 (Analysis and Design of Feedback Control Systems) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
3. (4 points) Prove the fundamental theorem concerning State Space in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
4. (6 points) Construct an original proof-level problem involving LQG Control for 2.14 (Analysis and Design of Feedback Control Systems) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
5. (4 points) Construct an original proof-level problem involving Robust Control for 2.14 (Analysis and Design of Feedback Control Systems) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
6. (3 points) Analyze the limitations of standard approaches to Nonlinear Control when applied to edge cases in 2.14. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (3 points) Prove the fundamental theorem concerning Transfer Functions in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
8. (3 points) Construct an original proof-level problem involving PID Control for 2.14 (Analysis and Design of Feedback Control Systems) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

9. (2 points) Analyze the limitations of standard approaches to State Space when applied to edge cases in 2.14. Construct explicit counterexamples showing where intuitive reasoning fails.
10. (5 points) Analyze the limitations of standard approaches to LQG Control when applied to edge cases in 2.14. Construct explicit counterexamples showing where intuitive reasoning fails.
11. (5 points) Prove the fundamental theorem concerning Robust Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (3 points) Prove the fundamental theorem concerning Nonlinear Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
13. (3 points) Construct an original proof-level problem involving Transfer Functions for 2.14 (Analysis and Design of Feedback Control Systems) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
14. (4 points) Prove the fundamental theorem concerning PID Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
15. (4 points) Prove the fundamental theorem concerning State Space in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
16. (4 points) Prove the fundamental theorem concerning LQG Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
17. (3 points) Prove the fundamental theorem concerning Robust Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
18. (7 points) Prove the fundamental theorem concerning Nonlinear Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (5 points) Analyze the limitations of standard approaches to Transfer Functions when applied to edge cases in 2.14. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (5 points) Construct an original proof-level problem involving PID Control for 2.14 (Analysis and Design of Feedback Control Systems) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
21. (3 points) Prove the fundamental theorem concerning State Space in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
22. (5 points) Prove the fundamental theorem concerning LQG Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (3 points) Prove the fundamental theorem concerning Robust Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
24. (4 points) Prove the fundamental theorem concerning Nonlinear Control in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
25. (4 points) Prove the fundamental theorem concerning Transfer Functions in the context of 2.14 (Analysis and Design of Feedback Control Systems). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

Solutions

Solution 1:

Solution approach: First recall the definitions and key theorems related to Transfer Functions from 2.14. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 2:

Solution approach: First recall the definitions and key theorems related to PID Control from 2.14. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 3:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 4:

Solution approach: First recall the definitions and key theorems related to LQG Control from 2.14. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 5:

Solution approach: First recall the definitions and key theorems related to Robust Control from 2.14. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Nonlinear Control. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 8:

Solution approach: First recall the definitions and key theorems related to PID Control from 2.14. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 9:

The edge case analysis reveals the subtle assumptions underlying standard treatments of State Space. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 10:

The edge case analysis reveals the subtle assumptions underlying standard treatments of LQG Control. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 11:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 13:

Solution approach: First recall the definitions and key theorems related to Transfer Functions from 2.14. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 14:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 15:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 16:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 17:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 18:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 19:

The edge case analysis reveals the subtle assumptions underlying standard treatments

of Transfer Functions. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

Solution approach: First recall the definitions and key theorems related to PID Control from 2.14. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 21:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 22:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 23:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 24:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 25:

The proof follows from the definitions and properties established in 2.14. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.